

Assignment
Data management & visual insight
Sem - 5

Assignment Instructions

1. All students **must complete all the sums** with full steps and clear explanations.
2. Your work should be **neat and clean** — present every solution properly.
3. After completing the assignment, **create a PDF** containing all the solved sums.
4. Submit the PDF to the **given Google submission link**.
5. **Last date for submission: 25th November**
Assignments submitted on or before the due date will be accepted.
Assignments submitted after the due date **will not be accepted**.

Question

1. A company produces rechargeable batteries. A random sample of **36 batteries** is tested, and the **average lifetime** is found to be **82 hours**. The **population standard deviation (σ)** is known to be **6 hours**.

Find:

- The **point estimate** of the population mean.
- The **standard error** of the mean.
- The **margin of error** for a 95% confidence level.
- The **95% confidence interval** for the population mean battery life.

2. A survey of **64 households** reveals that the **average monthly electricity bill** is **\$120**. The **population standard deviation** is **\$20**.

Find:

- Point estimate of the population mean monthly bill.
- Standard error of the mean.
- Margin of error for a **90% confidence level**.
- Construct the **90% confidence interval** for the true mean monthly electricity bill.

3. A coffee shop owner wants to estimate how much coffee customers drink on average per day. A random sample of **100 customers** shows an average consumption of **2.8 cups per day**. The **population standard deviation** is **0.5 cups**.

Find:

- Point estimate of the population mean daily coffee consumption.
- Standard error of the mean.
- Margin of error for a **99% confidence level**.
- The **99% confidence interval** for the true mean daily coffee consumption.

4. A sample of **25 employees** from a large firm reports a **mean daily commute time** of **42 minutes** with a **sample standard deviation** of **7 minutes**.

Find:

- Point estimate of the population mean commute time.
- Standard error of the mean.

- Margin of error for a **90% confidence level**.
- 90% confidence interval for the population mean commute time.

5. In a poll of **1,000 registered voters**, **620** say they support a new policy.

Find:

- Point estimate of the proportion of voters supporting the policy.
- Standard error of the proportion.
- Margin of error for a **90% confidence level**.
- Construct the **90% confidence interval** for the true proportion of support.

6. A factory inspects a sample of **400 bulbs** and finds **28 defective** ones.

Find:

- Point estimate of the proportion of defective bulbs.
- Standard error of the sample proportion.
- Margin of error for a **99% confidence level**.
- 99% confidence interval for the population proportion of defective bulbs.

7. Out of **500 applicants**, **315** were accepted into a college program.

Find:

- Point estimate of the admission proportion.
- Standard error of the proportion.
- Margin of error for a **95% confidence level**.
- Construct the **95% confidence interval** for the true admission proportion.

8. The average score on a national exam is **75**. A teacher believes her students score **lower than average**.

A sample of **20 students** shows a **mean score of 72** with a **sample standard deviation of 6**.

At $\alpha = 0.05$, test the teacher's claim.

9. A nutritionist claims that the average daily sodium intake for adults is **2,300 mg**.

A sample of **25 adults** shows a **mean intake of 2,420 mg** and a **sample standard deviation of 300 mg**.

At $\alpha = 0.01$, test whether the **average sodium intake differs** from the recommended value.

10. A machine fills cereal boxes that should contain **500 g** on average.

A sample of **100 boxes** has a **mean weight of 505 g**, and $\sigma = 8$ g.

At $\alpha = 0.05$, test whether the machine is **overfilling** the boxes.

11. A survey found that **60%** of customers prefer brand A.

A rival company wants to test if the proportion has **changed**.

In a new sample of **400 customers**, **250** preferred brand A.

At $\alpha = 0.05$, test whether the **true proportion differs from 0.60**.

12. A college claims that at least **70%** of its graduates find jobs within 3 months.
A survey of **120 graduates** shows **78** found jobs within that time.
At $\alpha = 0.05$, test whether **fewer than 70%** find jobs (i.e., test if the claim is overstated).
13. A website claims that **90%** of users are satisfied.
In a sample of **200 users**, **170** say they are satisfied.
At $\alpha = 0.01$, test whether the **satisfaction rate is below 90%**.
14. A nutrition bar company claims that the **average protein content** of their bars is **20 g**.
A sample of **16 bars** shows a **mean of 19.2 g** and a **sample standard deviation of 1.4 g**.
At $\alpha = 0.05$, test whether the **mean protein content differs from 20 g**.
15. A delivery service claims the **average delivery time** is **30 minutes**.
A random sample of **64 deliveries** shows a **mean of 32 minutes**, with a **population standard deviation of 6 minutes**.
At $\alpha = 0.05$, test whether the **average delivery time is greater than 30 minutes**.
16. A machine is designed to fill bottles with **500 ml** of juice.
A random sample of **40 bottles** has the following fill amounts (in ml):
498, 503, 501, 499, 502, 504, 497, 500, 503, 499,
501, 505, 496, 500, 502, 499, 504, 498, 503, 500,
497, 505, 502, 498, 504, 499, 501, 500, 497, 503,
499, 502, 500, 503, 497, 505, 504, 498, 501, 500
Population standard deviation: $\sigma = 4 \text{ ml}$
Tasks:
- Compute sample mean ($\bar{x}$).
 - Formulate hypotheses to test if the machine fills correctly ($\mu = 500$).
 - Compute standard error, margin of error (95% CI), and interval estimate.
 - Perform hypothesis test using **p-value** and **critical value** (Z-test).
 - Interpret results.
17. A company claims its energy drink contains **80 mg of caffeine** on average.
A random sample of **30 bottles** gives these caffeine contents (mg):
78, 81, 79, 80, 83, 77, 82, 79, 81, 84, 80, 82, 79, 78, 83, 77, 82, 79, 81, 80, 84, 83, 78, 79, 82,
80, 81, 79, 82, 83
 $\sigma = 3 \text{ mg}$
Test at $\alpha = 0.05$ if the mean caffeine content is **greater than 80 mg**.
Also find 95% CI, SE, and ME.

18. A dietitian believes that the average sodium content of a soup is **too high** compared to **650 mg**.
A sample of **10 cans** gives the following sodium contents (mg):
660, 645, 668, 652, 649, 655, 671, 648, 656, 659
Find:
- Sample mean and sample standard deviation.
 - SE, ME, and 95% CI.

- Develop hypotheses ($H_0: \mu = 650$).
- Perform **test** using both **p-value** and **t-critical**.

19. A class's average test score is claimed to be **75**.

A teacher thinks the score is **lower**.

A sample of **8 students** scores:

72, 70, 74, 76, 71, 73, 69, 75

Test at $\alpha = 0.05$ using the **t-test**.

Also find SE, ME, and 95% CI.

20. A car manufacturer claims its model accelerates from 0–60 mph in **7.5 seconds**.

A sample of **50 cars** gives a mean of **7.3 seconds**, with $\sigma = 0.5$ seconds.

- Test if cars accelerate **faster than claimed** ($\mu < 7.5$).
- Use both **p-value** and **critical Z**.
- Find 99% confidence interval.

21. An instructor believes the **average GPA** of students in her department is **3.0**.

A random sample of **12 students** gives GPAs:

3.1, 2.8, 3.3, 3.0, 3.2, 2.9, 3.4, 3.1, 2.7, 3.0, 3.2, 3.1

Test if the mean GPA differs from 3.0 ($\alpha = 0.05$).

Compute SE, ME, CI, and perform **t-test**.

22. A manufacturer claims **90%** of its devices pass quality tests.

In a sample of **200 devices**, **174 passed**.

Test at $\alpha = 0.05$ if the proportion differs from 0.90.

Find SE, ME, and 95% CI.

23. A marketing campaign claims that at least **40%** of customers prefer a new product.

Out of **250 surveyed**, **115 preferred it**.

Test at $\alpha = 0.05$ if the proportion exceeds 0.40.

Also find 95% CI, SE, and ME.

24. A beverage company compares average sugar content (grams) of two recipes A and B.

Two independent large production lines were sampled.

- Sample A: ($n_1 = 50$), ($\bar{x}_1 = 12.4$) g. Population SD ($\sigma_1 = 2.0$) g.
- Sample B: ($n_2 = 60$), ($\bar{x}_2 = 11.7$) g. Population SD ($\sigma_2 = 2.5$) g.

Test at ($\alpha = 0.05$) whether the mean sugar contents **differ** (two-tailed). Also produce the **95% confidence interval** for ($\mu_1 - \mu_2$).

Tasks:

- Compute the sample difference ($\bar{x}_1 - \bar{x}_2$).
- State (H_0) and (H_a).
- Compute SE for ($\bar{x}_1 - \bar{x}_2$) using known σ s.
- Compute margin of error and 95% CI.
- Find test statistic (Z), p-value, and compare with critical Z.
- State conclusion.

25. A machine manufacturer claims a new process increases average tensile strength. Two independent processes are compared.

- Process Old: ($n_1=40$), ($\bar{x}_1=195$) MPa, ($\sigma_1=6$) MPa.
- Process New: ($n_2=35$), ($\bar{x}_2=199$) MPa, ($\sigma_1=7$) MPa.

Test at ($\alpha=0.01$) whether the new process yields **higher mean strength** ($H_a: \mu_1 > \mu_2$)).

Provide the **99% CI** for ($\mu_1 - \mu_2$).

Tasks: same checklist (use right-tailed Z test).

26. A nutrition study compares average daily protein intake for men and women in a sample.

- Men: ($n_1=28$), ($\bar{x}_1=75$) g, ($s_1=12$) g.
- Women: ($n_2=32$), ($\bar{x}_1=68$) g, ($s_2=10$) g.

Use ($\alpha=0.05$) to test whether mean protein intake differs between sexes. Use the appropriate **Welch t-test** (unequal variances). Also compute **95% CI** for ($\mu_1 - \mu_2$).

Tasks:

1. Compute ($\bar{x}_1 - \bar{x}_2$).
2. State hypotheses (two-tailed).
3. Compute SE for difference using (s_1, s_2).
4. Compute degrees of freedom (Welch approximation), t-critical, margin, CI.
5. Compute t statistic, p-value, and compare with critical t.
6. Conclude.

27. A pharmaceutical trial compares mean blood pressure reduction for two drugs. Variances can be assumed equal.

- Drug A: ($n_1=22$), ($\bar{x}_1=12.5$) mmHg, ($s_1=3.1$).
- Drug B: ($n_2=20$), ($\bar{x}_2=10.8$) mmHg, ($s_2=2.9$).

Test at ($\alpha=0.05$) whether the mean reductions differ. Use **pooled t-test** and give a **95% CI** for ($\mu_1 - \mu_2$).

Tasks: compute pooled variance, SE (pooled), t-statistic, df, p-value, critical t, ME, CI.

28. A company tests two ad designs and records whether viewers clicked (success).

- Ad A: ($n_1=800$), clicks ($\bar{x}_1=176$).
- Ad B: ($n_2=750$), clicks ($\bar{x}_2=150$).

Test at ($\alpha=0.05$) whether the click-through proportions differ ($H_0: p_1=p_2$)). Also produce the **95% CI** for ($p_1 - p_2$).

Tasks:

1. Compute ($\bar{p}_1, \bar{p}_2$) and pooled ($\bar{p}$).
2. Compute SE (pooled) for hypothesis test and unpooled SE for CI.
3. Compute Z statistic, p-value, critical Z, ME, and CI.
4. Interpret.

29. A hospital claims new follow-up calls reduce readmission proportion below the current 12%. A pilot compares old vs new program.

- Old program: ($n_1=400$), readmissions ($\bar{x}_1=52$).
- New program: ($n_2=420$), readmissions ($\bar{x}_2=36$).

Test at ($\alpha=0.01$) whether the new program has a **lower** readmission proportion ($H_a: p_1 < p_2$)). Also find the **99% CI** for ($p_1 - p_2$).

Tasks: pooled test for p difference (left-tailed), p-value and critical Z, plus CI.

30. A supermarket trial tests whether a new display increases the proportion of customers buying an add-on product.

- Control: ($n_1=500$), purchases ($x_1=65$).
- Display: ($n_2=520$), purchases ($x_2=88$).

Test at ($\alpha=0.05$) whether the new display increases the purchase proportion (($H_a: p_1 < p_2$)). Also compute **95% CI** for ($p_1 - p_2$).

Tasks: compute ($\bar{p}$), pooled test, Z, p-value (right-tailed), critical Z, ME, CI.

31. A snack company believes customer preferences for its snacks follow these proportions:

Snack Type	Claimed Proportion
Chips	0.40
Cookies	0.30
Nuts	0.20
Popcorn	0.10

In a survey of **200** customers, the results are:

Snack Type	Observed
Chips	90
Cookies	65
Nuts	30
Popcorn	15

Test at $\alpha = 0.05$ whether customer preferences follow the claimed distribution.

32. A car dealer claims that the color preferences for buying a new car are:

Color	Theoretical %
White	25%
Black	30%
Gray	20%
Blue	15%
Other	10%

A sample of **300** new car buyers shows:

Color	Observed
White	65
Black	90
Gray	70
Blue	40
Other	35

Test the goodness-of-fit at $\alpha = 0.01$.

33. An online learning site expects enrollment distribution in course categories as:

Category	Expected Proportion
Technology	0.25
Business	0.20
Arts	0.10
Language	0.15
Health	0.20
Other	0.10

From **400** enrollments, the counts are:

Category	Observed
Technology	110
Business	70
Arts	30
Language	55
Health	90
Other	45

Test the claim using $\alpha = 0.05$.

34. Gender vs. Music Preference

A survey records preferred music genre by gender:

	Pop	Rock	Classical	Total
Male	30	42	18	90
Female	50	30	30	110
Total	80	72	48	200

Test at $\alpha = 0.05$:

Is music preference independent of gender?

35. Smoking Habit vs. Exercise Frequency

Data collected from 300 adults:

	Regular Exercise	Occasional Exercise	No Exercise	Total
Smoker	20	45	35	100
Non-Smoker	80	90	30	200
Total	100	135	65	300

Test at $\alpha = 0.01$:

Are smoking and exercise habits independent?

36. Employee Department vs. Job Satisfaction

Survey responses:

Department	Satisfied	Neutral	Dissatisfied	Total
HR	25	10	15	50
Sales	40	25	35	100
IT	35	20	15	70
Total	100	55	65	220

Test at $\alpha = 0.05$:

Is job satisfaction independent of department?

37. A poll asks whether voters support, oppose, or are neutral toward a new policy:

Political Party	Support	Oppose	Neutral	Total
Party A	60	35	25	120
Party B	45	55	30	130
Party C	30	25	15	70
Total	135	115	70	320

Test at $\alpha = 0.05$:

Are policy opinions independent of political affiliation?

38. Training Methods

A manager wants to test whether three different employee training programs lead to different productivity scores.

Group	Sample Size	Mean Score	Standard Deviation
Method A	14	82	5
Method B	12	88	6
Method C	10	91	7

Test at $\alpha = 0.05$:

Are the mean productivity scores significantly different among training methods?

Create ANOVA Table

39. Fertilizer Types

A botanist tests three fertilizers on plant height (cm). Data collected:

- Fertilizer A: 42, 45, 40, 47, 43
- Fertilizer B: 50, 55, 52, 53, 56
- Fertilizer C: 48, 45, 47, 49, 46

Test at $\alpha = 0.01$ if the fertilizers produce different average plant heights.

Create ANOVA Table

40. Diet Plans and Weight Loss

A study tests four diet plans measuring weight loss over 6 weeks:

Diet Plan	Sample Size	Mean Loss (kg)	Std. Dev.
Diet A	20	4.2	1.1
Diet B	18	5.0	1.3
Diet C	22	6.1	1.0
Diet D	16	4.8	1.4

Use $\alpha = 0.05$:

Are the mean weight losses different across diets?

Create ANOVA Table

41. Student Exam Performance by Teaching Style

A teacher tries 3 teaching approaches:

- Approach 1: 67, 72, 70, 65, 68, 74
- Approach 2: 75, 78, 82, 79, 77
- Approach 3: 60, 62, 58, 65, 61, 59

At $\alpha = 0.05$, determine if teaching style affects exam performance.

Create ANOVA Table

42. Average Wait Times at Restaurants

A researcher compares wait times (minutes) at 4 restaurants:

Restaurant	Observed Wait Times
R1	18, 20, 22, 19, 21
R2	25, 28, 30, 27, 26
R3	15, 17, 14, 16, 18
R4	20, 22, 19, 21, 23

Test at $\alpha = 0.05$:

Do the restaurants have different mean wait times?

Create ANOVA Table

43. Fertilizer Efficiency

4 fertilizers tested in 4 plots, yield (kg):

Plot	F1	F2	F3	F4
1	20	22	24	23
2	21	23	25	22
3	19	21	23	21
4	20	22	24	23

Test at $\alpha = 0.05$

Create ANOVA Table

44. Training Methods

3 methods tested on 5 instructors for performance rating:

Instructor	T1	T2	T3
I1	78	82	80
I2	75	81	79
I3	77	83	82
I4	80	84	81
I5	76	82	80

Test at $\alpha = 0.05$

Create ANOVA Table

45. Battery Life

Factors:

A = Brand (A1, A2)

B = Temperature (Low, High)

Replications = 3

Brand	Low	Low	Low	High	High	High
A1	18	19	17	14	15	13
A2	20	21	19	17	16	18

Test at $\alpha = 0.05$

Create ANOVA Table

46. Crop Growth

A = Fertilizer (F1, F2, F3)

B = Irrigation (I1, I2)

Replications = 2

Fertilizer	I1	I1	I2	I2
F1	25	27	30	29
F2	28	30	32	31
F3	26	25	29	27

Test at $\alpha = 0.01$

Create ANOVA Table

47. Packaging $\times$ Display

A = Packaging Type (Box, Bottle)

B = Store Display (Front, Side, Back)

Replications = 3

Sales Units (raw data):

Packaging	Front	Side	Back
Box	120,122,118	110,112,109	105,107,106
Bottle	130,131,128	125,124,123	110,112,111

Test at $\alpha = 0.05$

Create ANOVA Table

48. Fuel Type $\times$ Engine Design

A = Fuel Type (Gasoline, Ethanol, Hybrid)

B = Engine Design (Design 1, Design 2)

Replications = 3

Fuel Type	D1 Data	D2 Data
Gasoline	210, 205, 208	200, 198, 202
Ethanol	195, 198, 194	185, 182, 187
Hybrid	225, 230, 228	220, 218, 223

Test at $\alpha = 0.01$

Create ANOVA Table

49. Quality Inspection

A quality engineer records whether an item produced each hour is Good (G) or Defective (D):

Sequence (20 items):

G, G, D, G, D, D, G, G, G, D, G, D, D, D, G, G, D, G, D, G

Test ($\alpha = 0.05$): Are the data random?

50. Stock Market Movement

Daily stock price change:

+ = Increase, - = Decrease

Sequence (18 days):

+, -, -, +, +, -, -, -, +, +, +, -, +, -, -, +, +, -

Test randomness at $\alpha = 0.05$.

51. Machine Output

Classify output above or below median (B = Below, A = Above):

Sequence (15 outputs):

B, A, A, B, B, A, B, A, A, A, B, B, A, B, B

Is the sequence random? Use $\alpha = 0.10$.

52. Customer Arrivals

“L” = Low arrivals, “H” = High arrivals (based on threshold):

L, H, L, L, H, H, H, L, H, H, L, L, L, H

Test randomness at $\alpha = 0.05$.

53. Website Traffic

Classified relative to average:

H, H, H, L, L, H, L, H, H, L, L, L, H, H, L, H

Test at $\alpha = 0.05$ whether highs and lows are random.

54. Teaching Method Effect

Test scores before vs after new teaching method:

Differences (After – Before):

3, 5, 4, -1, 2, -2, 3, 1, 4, -1

Test at $\alpha = 0.05$ whether method improves scores (one-tailed).

55. Pain Reduction

Pain scores (lower is better):

Before: 7, 8, 6, 7, 9, 8

After: 5, 7, 6, 6, 7, 8

At $\alpha = 0.10$, test whether treatment reduces pain (one-tailed).

56. Median Comparison

Service response times vs a claimed median of 15 minutes:

Sample data:

16, 14, 15, 13, 18, 17, 14, 16, 12, 19

Test ($\alpha = 0.05$, two-tailed) if the median is different from 15.

57. Machine Calibration

Paired measurements: Standard vs Machine output

Difference (Machine – Standard):

1, 0, -2, -1, 1, 2, 0, -1, -2, 1

Test ($\alpha = 0.05$, two-tailed): Does the machine produce a median difference of 0?

58. WSR-1: Meditation & Stress

A researcher measures stress scores before and after a 2-week meditation program:

| Before | 18 | 21 | 23 | 20 | 19 | 24 | 22 | 25 |

| After | 16 | 18 | 21 | 19 | 17 | 22 | 21 | 23 |

Test at $\alpha = 0.05$: Does meditation reduce stress?

59. WSR-2: Training Program for Typing Speed

Typing speed (wpm) before & after training:

| Before | 52 | 55 | 60 | 57 | 58 | 54 | 59 |

| After | 56 | 58 | 63 | 60 | 61 | 57 | 62 |

Test if training improves typing speed (one-tailed, $\alpha = 0.01$)

60. WSR-3: Pain Treatment on Joint Pain

Pain scale: higher = more pain

| Before | 9 | 7 | 8 | 10 | 9 | 6 |

| After | 6 | 6 | 7 | 8 | 7 | 5 |

Test at $\alpha = 0.05$ whether treatment reduces pain.

61. WSR-4: Productivity Before & After Workplace Lighting Change

| Before | 42 | 45 | 44 | 46 | 47 | 43 | 48 | 44 |

| After | 44 | 46 | 46 | 47 | 49 | 45 | 50 | 47 |

Test if new lighting improves productivity (two-tailed)

62. MW-1: Smartphone Battery Life

Two brands tested (hours):

Brand A: 10, 12, 9, 11, 13, 10

Brand B: 14, 13, 15, 12, 16, 14

Test at $\alpha = 0.05$ if mean life differs.

63. MW-2: Customer Satisfaction by Two Stores

Rating (1–10):

Store A: 8, 7, 9, 6, 8

Store B: 6, 5, 7, 6, 7

One-tailed test at $\alpha = 0.05$: Store A higher satisfaction?

64. MW-3: Study Strategy Comparison

Test scores:

Strategy A: 75, 78, 80, 74, 76

Strategy B: 82, 85, 81, 83, 84

Test $\alpha = 0.01$, any performance difference?

65. MW-4: Reaction Times for Two Game Controllers

Time (ms):

Controller X: 220, 215, 230, 225, 218

Controller Y: 205, 200, 210, 208, 207

At $\alpha = 0.05$, is controller Y faster?

66. KW-1: Employee Engagement by Department

Engagement scores:

HR: 72, 75, 71, 73

Sales: 78, 80, 79, 77

IT: 70, 68, 69, 71

Test at $\alpha = 0.05$: Same medians?

67. KW-2: Crop Yield under 3 Fertilizers

Yield:

F1: 20, 22, 21, 23

F2: 25, 24, 26, 27

F3: 18, 17, 19, 18

Test at $\alpha = 0.01$.

68. KW-3: Delivery Time for Different Courier Services

Delivery (days):

C1: 3, 4, 3, 2, 4

C2: 5, 6, 5, 4, 5

C3: 4, 4, 3, 5, 4

Test $\alpha = 0.05$.

69. RC-1: IQ vs Academic Score

10 students ranked in IQ and exam score:

IQ Rank: 3, 1, 4, 2, 5, 8, 6, 9, 7, 10

Exam Rank: 2, 1, 3, 4, 5, 7, 6, 8, 9, 10

Test correlation at $\alpha = 0.05$.

70. RC-2: Social Media Followers vs Content Quality

12 influencers:

Followers Rank: 1-12

Quality Rank: 3,1,4,2,5,7,6,9,8,11,10,12

Test positive correlation ($\alpha = 0.01$, one-tailed)